Sebastián José Cardona Ramírez

Backend Developer | REST API Design & Construction

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PROFESSIONAL PROFILE

Software Engineer specializing in backend development and REST API design, with a solid foundation in data modeling and authentication/authorization, and a growing focus on AI automation and the Semantic Web. I build production-oriented systems: real-time IoT data pipelines, secure distributed architectures, and complex relational modeling, confidently leading end-to-end backend development from database design to cloud deployment.

TECHNICAL SKILLS

Languages:	Python, JavaScript, SQL, C#
Backend & Frameworks:	FastAPI, Django, Express.js, ASP.NET, Node.js
APIs & Security:	REST APIs, SOAP/WSDL, JWT (Asymmetric RS256), OAuth2, Argon2, Refresh Tokens, RBAC, Hardening, AES-256
Databases & Caching:	PostgreSQL, MySQL, MongoDB, Redis, Supabase, SQLAlchemy, Alembic (migrations)
Semantic Web & AI:	OWL 2 Ontologies, Protégé, HermiT Reasoner, Owlready2, Apache Jena, LEGAL-BETO (NLP), SWRL Rules, LLM applications
IoT & Messaging:	MQTT, HiveMQ, WebSockets, Arduino IDE
DevOps & Cloud:	Docker, Docker Compose, Git/GitHub, Basic CI, Render, Vercel, Replit

KEY PROJECTS

LawSim Pymes – Semantic Architecture & Regulatory Simulation Platform — 2026

Vue.js, Node.js, Express, Python, Owlready2, Apache Jena, HermiT Reasoner, LEGAL-BETO (NLP), PostgreSQL, Redis, SOAP/WSDL

- Architected the technical blueprint and hybrid topology (Layered + Microservices + MVC) to decouple standard CRUD operations from heavy logic processing pipelines for corporate compliance.
- Modeled an OWL 2 domain ontology in Protégé and integrated the HermiT Reasoner alongside the LEGAL-BETO (NLP) language model to ingest dynamic Colombian regulations and infer operational risks via SWRL rules.
- Engineered formalized SOAP/WSDL microservice interfaces (NormaService and SimulacionService) and enforced strict cryptographic security guardrails with asymmetric JWT tokens (RS256) and AES-256 at-rest data encryption.
- Mapped intuitive interactive user wizards and predictable navigation structures using hypermedia web engineering methodologies (UWE and OOHDM).

Industrial Environmental Monitoring Platform — 2025

Python, FastAPI, PostgreSQL (Supabase), MQTT/HiveMQ, JWT, Argon2, WebSockets, Docker, Render

- Designed and built a real-time monitoring backend that ingests temperature, humidity, and CO2 telemetry from ESP32 sensors via MQTT, persisting it in PostgreSQL for historical analytics.
- Implemented a secure REST API with JWT authentication, password hashing with Argon2, and Role-Based Access Control (RBAC: admin, operator, viewer).
- Built a live monitoring dashboard using WebSockets alongside configurable data validation rules with range boundaries to instantly flag risk conditions in industrial fermentation lines.
- Deployed as a monolith on Render backed by a managed PostgreSQL instance (Supabase); configured a local development environment fully containerized with Docker Compose.

Car Inspection API – Fleet Management Inspection & Reporting Backend **2025 (In progress)**

Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, JWT (Access + Opaque Refresh tokens), Docker

- Designed a relational data model for vehicles, inspection logs, multimedia evidence, digital authorizations, and immutable audit footprints, tracking schema states with versioned migrations via Alembic.
- Implemented authentication using short-lived JWT access tokens and rotating opaque refresh tokens, incorporating strict rate-limiting policies on login endpoints to mitigate brute-force attacks.
- Built robust endpoints for vehicle onboarding, comprehensive inspection reporting (findings, severity levels, final outcomes), and file upload architectures for photo/audio evidence.
- Modeled a complete, transparent historical log (storing old vs. new values) tracking every administrative state change performed over active inspection reports.

EDUCATION

Software Engineering – Universidad Cooperativa de Colombia **2022 - 2026**

LANGUAGES

Spanish: Native | **English:** B1